

Feedback — Module 4 Quiz **No Grace Period**

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You submitted this quiz on **Sun 4 Oct 2015 10:53 PM PDT**. You got a score of **5.00** out of **5.00**.

Question 1

When performing gene set analysis it is critical to use the same annotation as was used in pre-processing steps. Read the paper behind the Bottomly data set on the ReCount database:

<http://www.ncbi.nlm.nih.gov/pubmed?term=21455293> Using the paper and the function:

`supportedGenomes()` in the `goseq` package can you figure out which of the Mouse genome builds they aligned the reads to.

Your Answer	Score	Explanation
☐ UCSC hg19		
☐ Genome Reference Consortium GRCm38		
☒ UCSC mm9	✓ 1.00	
☐ UCSC hg18		
Total	1.00 / 1.00	

Question 2

Load the Bottomly data with the following code and perform a differential expression analysis using `limma` with only the strain variable as an outcome. How many genes are differentially expressed at the 5% FDR level using Benjamini-Hochberg correction? What is the gene identifier of the first gene differentially expressed at this level (just in order, not the smallest FDR) ? (hint: the `featureNames` function may be useful)

```
library(Biobase)
library(limma)
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
```

```

bot = bottomly.eset
pdata_bot=pData(bot)
fdata_bot = featureData(bot)
edata = exprs(bot)
fdata_bot = fdata_bot[rowMeans(edata) > 5]
edata = edata[rowMeans(edata) > 5, ]

```

Your Answer**Score****Explanation**
☐ 223 at FDR 5%; ENSMUSG000000027855 first DE gene

☐ 223 at FDR 5%; ENSMUSG000000000001 first DE gene

☐ 9431 at FDR 5%; ENSMUSG000000027855 first DE gene

☒ 223 at FDR 5%; ENSMUSG000000000402 first DE gene  1.00

Total

1.00 / 1.00

Question 3

Use the `nullp` and `goseq` functions in the `goseq` package to perform a gene ontology analysis. What is the top category that comes up as over represented? (hint: you will need to use the genome information on the genome from question 1 and the differential expression analysis from question 2.

Your Answer**Score****Explanation**
☐ GO:0038023

☐ GO:0001653

☒ GO:0004888  1.00

☐ GO:0008528

Total

1.00 / 1.00

Question 4

Look up the GO category that was the top category from the previous question. What is the

name of the category?

Your Answer	Score	Explanation
☐ detection of chemical stimulus involved in sensory perception of sour taste		
☒ transmembrane signaling receptor activity	✓ 1.00	
☐ intracellular membrane-bounded organelle		
☐ signaling receptor activity		
Total	1.00 / 1.00	

Question 5

Load the Bottomly data with the following code and perform a differential expression analysis using `limma` and treating strain as the outcome but adjusting for lane as a factor. Then find genes significant at the 5% FDR rate using the Benjamini Hochberg correction and perform the gene set analysis with `goseq` following the protocol from the first 4 questions. How many of the top 10 overrepresented categories are the same for the adjusted and unadjusted analysis?

```
library(Biobase)
library(limma)
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata_bot=pData(bot)
fdata_bot = featureData(bot)
edata = exprs(bot)
fdata_bot = fdata_bot[rowMeans(edata) > 5]
edata = edata[rowMeans(edata) > 5, ]
```

Your Answer	Score	Explanation
☒ 2	✓ 1.00	
☐ 10		
☐ 3		

0

Total

1.00 / 1.00